# science

# Performance and assessments answers

# 2nd prep

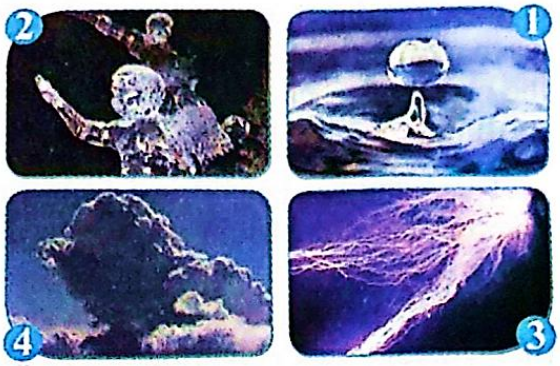
# Week 1, 2

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# I. Classroom Performance: States of Matter

## Study the following figure , then answer:

**1-Mention the name of each case?**

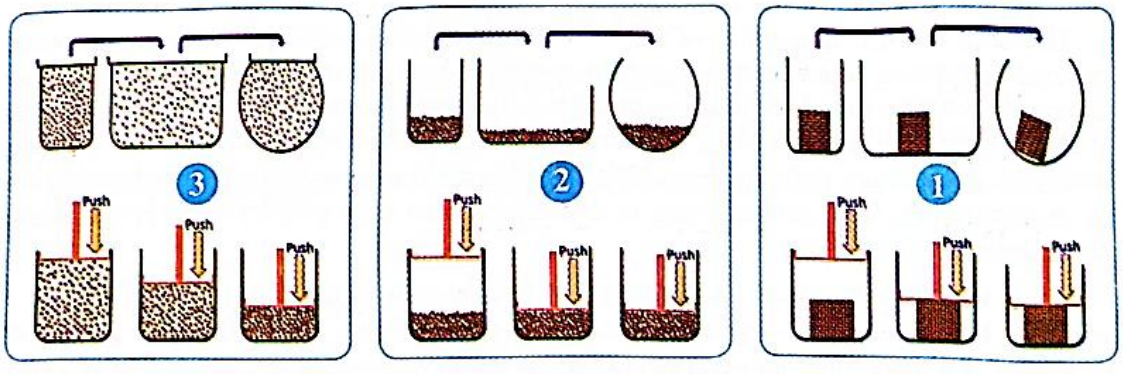



| Figure No. | Name of the State |
|---|---|
| 1 | Liquid State |
| 2 | Solid State |
| 3 | Plasma State |
| 4 | Gas state |

1. Which state of matter has a definite shape and volume?
   - **Answer: Solid State**
2. Which states are spillable?
   - **Answer: Liquid State** and **Gaseous State** (collectively called **Fluids**)
3. The state of matter that can **spread** is:
   - **Answer: Gaseous State**
4. Matter can be **compressed** when it is in the state of:
   - **Answer: Gaseous State**
5. Liquids and gases are called:
   - **Answer: Fluids**

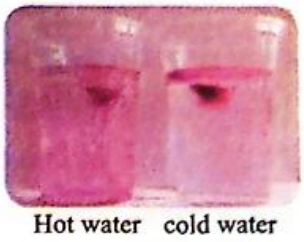
# II. Note and Identify: Matter Properties

## Study the following figures, which illustrate the properties of matter states, then identify the numbers of the images and state of matter that represent each of the following:



| Property | Figure No | State of Matter |
|---|---|---|
| 1. **Its shape** does not change **by changing the container.** | **(1)** | **Solid** |
| 2. **Its shape** changes **while its volume is** constant**.** | **(2)** | **Liquid** |
| 3. **Its shape and volume** can be changed**.** | **(3)** | **Gas** |
| 4. **States of matter that** can be compressed | **(3)** | **Gases** |
| 5. **States of matter that** cannot be compressed | **(1), (2)** | **Solids** and **Liquids** |

1. From the opposite figure, the color of Potassium Permanganate powder spreads faster in **hot water**. Why?
   - **Answer:** Increasing the temperature **increases the kinetic energy** of the water particles, which causes faster movement and leads to quicker diffusion of the Potassium Permanganate particles.




## IV. Particle Theory

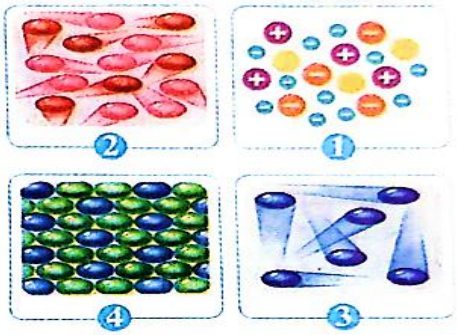
### Firstly:

Figure (1) represents the case of the **plasma state**

Figure (2) represents the case of the **liquid state**

Figure (3) represents the case of a **gas state**

Figure (4) represents the case of a **solid state**




### Secondly:

- The **interparticle spaces** are **as large as possible** in the case of the **Gas state**. And this is attributed to the **very weak attraction forces** between the particles.
- The **interparticle spaces** are **as small as possible** in the case of: **Solid**. This is attributed to the **very strong attraction forces** between the particles.

**-Using school book and based on what you have studied, explain the meaning of**

| | |
|---|---|
| Matter | **Anything that has** mass **and occupies space.** |
| Particles | tiny spheres that make up everything |
| Particle Theory | A theory that explains the behavior and properties of matter. |
| Plasma | The **fourth state of matter**, in which gases are ionized and converted into positively charged ions and negatively charged free electrons. |

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# Home performance

1. **What are the factors that determine the state of matter?**
   - **Answer:** The **attraction forces** between particles, the interparticle spaces between the particles, the way the particles can move and the **kinetic energy** of the particles .

**2- What are the factors that determine the speed of diffusion in fluids?**

- **Answer:** the state of matter and the temperature of the medium

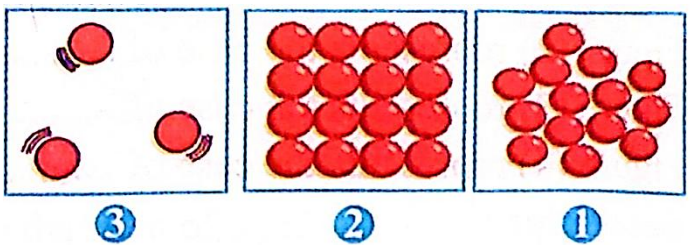
**3- Can you hold air? And why?**

- No, I can't hold air. Because air is a gas and gases have no fixed shape or volume due to the forces of attraction between particles are **very weak,** so gas particles move randomly and completely freely.

2. **Why does water spill when you tilt the cup?**
   - **Answer:** Because water is a **liquid**, the forces of attraction between its particles are relatively weak, allowing them to **slide over each other** (flow), and it takes the shape of its container.

3. **Why does a sugar cube not take the shape of the cup?**
   - **Answer:** Because the sugar cube is a **solid**, its particles are tightly packed and the forces of attraction between its particles are very strong, giving it a **definite shape and volume**.

**-Study the opposite figure well, and then answer:**




1- Substance No.**3**. can be compressed.
2- Substances.**1**.. and.**3**. are called fluids.
3- Substance..**2**. retains a constant volume and shape
4- The most widespread substance is **3** because **it is a gas and gases have very weak forces of attraction between its particles, so gas particles move randomly and completely freely.**

## Weekly assessment (1, 2)

**Write down what the following sentences mean:**

| | |
|---|---|
| 1. Anything that has mass and volume. | **Matter** |
| 2. A state of matter that has a definite shape and fixed volume. | **Solid State** |
| 3. A state of matter that has the ability to spread and be compressed. | **Gas State** |
| 4. A state of matter that has a fixed volume and varies in shape with the container that contains it. | **Liquid State** |
| 5. A theory that explains the behavior and properties of matter. | **Particle Theory** |
| 6- The movement of particles of matter in fluids is random in all directions. | **Brownian Motion** |
| 7-A state of matter that occurs when gases are ionized into positively and negatively charged ions. | **Plasma state** |

## VIII. Give a Scientific Explanation for each of the following

1. **Solid matter has a fixed shape and definite volume.**

- This is because the attractive forces between solid particles are very strong, and the spaces between them are very small.

**2. Gaseous matter has no fixed shape or definite volume.**

- This is because the attractive forces between their particles are very weak, and the spaces between them are very large.

**3- Liquid matter is flowable.**

- Because the attractive forces between their particles are relatively weak, so it has the ability to flow and take the shape of the container

**4- Gases are compressible.**

- Because its interparticle spaces are very large

**5- Particles of matter spread faster in gases than in liquids.**

- Because the attraction forces between the gas particles are much weaker than the attraction force in liquids, and the interparticle spaces between them are much larger than the interparticle spaces of liquids

**6- Plasma is used in air conditioners.**

- To improve the quality of the air in enclosed spaces. As gases are passed through a high electric field to convert their atoms into positively charged ions and negatively charged free electrons. Then the charged gas ions break down harmful air pollutants, germs, and viruses, which makes the air more pure.

**7- It is difficult to break a piece of rock.**

- This is because the attractive forces between the particles of a solid rock are very strong, and the spaces between them are very small, which makes the particles very tightly bound to each other.

8- **The possibility of dissolving a quantity of table salt in water.**

- This is because the attractive forces between the particles of liquid water are relatively weak, and the spaces between them are relatively large, allowing the salt particles to occupy them.

**9- Spreading of smoke produces from a candle wick in the air.**

- This is because the attractive forces between the particles of air are very weak, and the spaces between them are very large, so the smoke particles can diffuse into them.

# science

# Performance and assessments answers

# 2nd prep

# Week 3, 4

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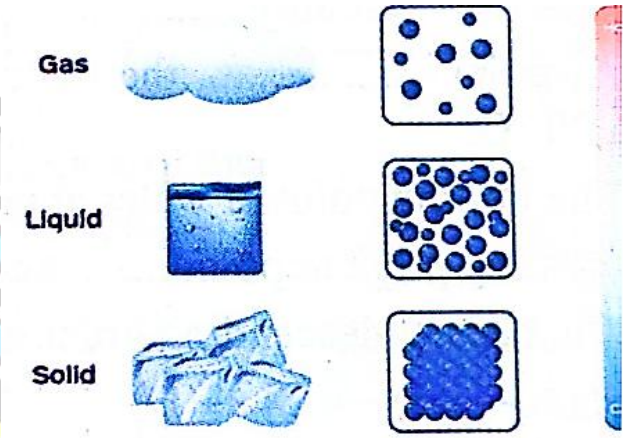
# Lesson (2)

# Changes in the state of matter

## Weeks 3, 4: Classroom performance

## Change in the physical state of a substance:

**Look at the figure in front of you and complete the following:**




1- Matter changes from one state to another, As shown in the figure, either by **heating** or **cooling** ( **gain of heat** or **loss of heat**).

2- When particles of matter acquire **thermal** energy, the Kinetic energy of particles increases, and the forces of attraction between them decrease, and the distances **between the particles** increase, so it transforms into **Liquid** or **Gas** state.

3- When the particles of matter lose **heat,** the **kinetic** energy of the particles decreases, and they become closer to each other; the attraction forces between them increase, and change to a liquid or solid state.

4- At normal atmospheric pressure, melting point of ice = 0◦C, while boiling point of water = 100◦C.

5- The process of **condensation** is the opposite of the process of **boiling**, and the process of freezing is the opposite of the process of **melting**.

6- The processes of **freezing** and **condensation** occur with a loss of heat energy, while the processes of **melting** and **boiling** occur with a gain of heat energy.

## Factors affecting the melting and boiling points

**Complete the following sentences:**

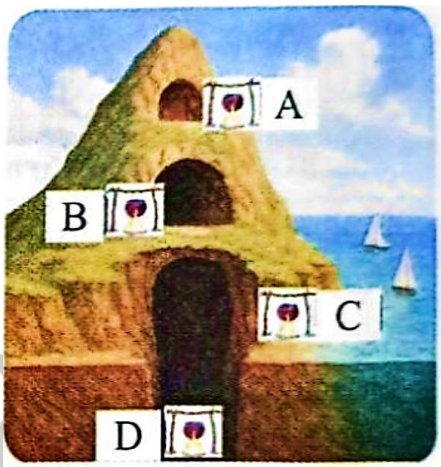
- Pure water boils at a temperature of **100°C,** and as atmospheric pressure increases, the boiling point of water increases above 100°C, while its freezing point decreases below 0°C.
- As we rise up above the surface of the Earth, atmospheric pressure **decreases** and boiling point of water becomes **lower** than 100°C.

## Pressure cooker (Steam pot)



- Pressure cookers **increase** the rate of cooking food by trapping **water vapor (steam)** and increasing pressure, causing water to boil at a higher temperature than in regular pots.

**Look at the figure in front of you and complete the following:**




- The highest boiling point of water is at the point **D** because **atmospheric pressure is highest there** (or **it is at the lowest altitude**).
- The lowest boiling point of water is at the point **A** because **atmospheric pressure is lowest there** (or **it is at the highest altitude**).
- At point **C**, the boiling point of water is 100°C.
- The boiling point of water at point (B) is higher than its boiling point at point **A** and lower than its boiling point at point **C**.
- The correct descending order of boiling point of water at the four points is: **(D) → (C) → (B) → (A)**
-

## Purity degree of matter

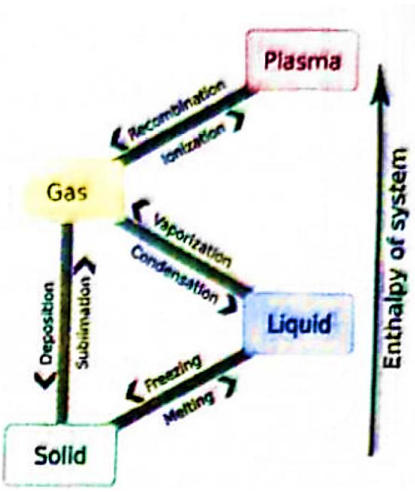
**Complete:**

- **Impurities** Affecting on the attraction forces between particles of pure substance, that changes the quantity of **energy** required to change its state, so both of **melting** and **boiling** points of that substance will change.

## Conversions of The Matter are Reversible Processes

**Study the figure in front of you, and then complete:**

- The matter undergoes a series of some reversible physical processes, those occur by **gaining** or **losing** thermal energy.
- Melting is the reverse of **freezing** process.
- **Condensation** is the reverse process of evaporation.
- **Sublimation** is the reverse process of deposition.
- The change of a substance from the solid state directly to the gaseous state without passing through the liquid state is known as **sublimation** as in **dry ice** and **solid iodine**, while the change of a substance from the gaseous state directly to the solid state without passing through the liquid state is known as **deposition**.

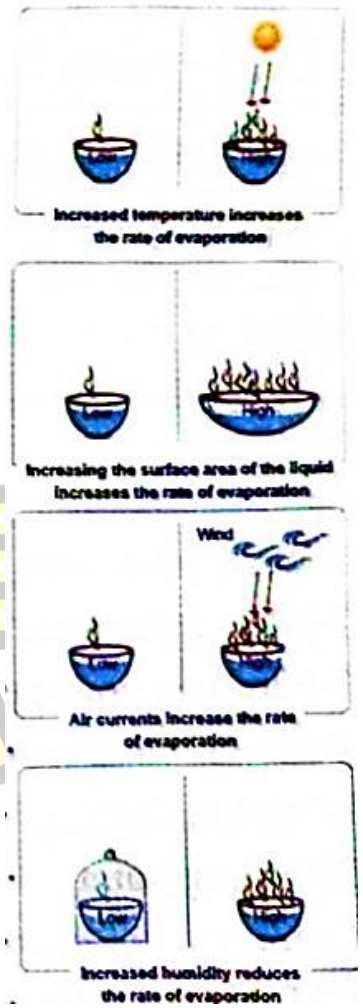


- Water vapour condenses when it loses **thermal energy** due to contact with a cold surface and turns into liquid droplets on plant leaves in the early morning.
  "Wet clothes dry when left exposed to the air without reaching boiling point." From the above statement, the degree of water evaporation is **below** its boiling point.

## Factors Affecting the Rate of Evaporation of Liquids

-There are a number of factors that affect the rate of evaporation of liquids.

- Evaporation occurs on the surface of the liquid at any temperature.
- For evaporation to occur, the molecules on the surface of the liquid must have enough energy to overcome the forces of attraction between the molecules of the liquid.




- **Study the figures in front of you and answer:**
- What factors affect the rate of evaporation? Explain the relation between them.

1. **Temperature:** The evaporation rate increases by increasing temperature. The evaporation rate increases due to an increase in the number of molecules that have sufficient thermal energy from the surrounding medium to overcome the attractive forces of other molecules and escape from the surface of the liquid into the air.
2. **Surface area of the liquid exposed:** The evaporation rate increases by increasing the surface area of the liquid which exposed to evaporation. The evaporation rate increases due to an increase in the number of molecules that can gain the required thermal energy from the surrounding medium to be released from the surface of the liquid
3. **Air Currents**: The evaporation rate increases with increasing air currents. Air currents increase the evaporation of more water molecules from the surface of the liquid
4. **Humidity:** The evaporation rate increases as decreasing humidity percentage. Trapping water vapor in the air of the container (1) reduces the evaporation rate, while releasing water vapor from container (2), which is uncovered, increases the rate of evaporation of more surface molecules

## Instant Coffee (quick dissolving)



- Coffee that dissolves in water **faster** than regular coffee.
- It is manufactured by exposing sprayed concentrated coffee to **hot, dry air;** its temperature is around 250 °C.
- The increase in the surface area of the spray coffee droplets exposed to hot air increases the rate of **evaporation** and the formation of solid instant coffee **crystals**.

**Assisted on the schoolbook and, in light of what you have studied, explain the meaning of the following words:**

- **Melting:** The process by which a solid substance gains a quantity of thermal energy either by heating or from the surrounding medium and converts to a liquid state
- **Freezing:** The process by which a solid substance loses a quantity of thermal energy by cooling and converts to a liquid state
- **Evaporation:** The process of conversion of matter from a liquid state to a gas state at any temperature less than its boiling point
- **Boiling:** The process by which a liquid substance gains a quantity of thermal energy by heating and converts to a gaseous state
- **Condensation:** The process by which a gas changes into a liquid, by losing thermal energy.
- **Deposition:** The changes of a substance directly from the gaseous state to the solid state without passing through the liquid state.
- **Sublimation:** The changes of a substance directly from the solid state to the gaseous state without passing through the liquid state.

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## Home Performance

**- Answer the Following**

**1-What factors impact melting and boiling points**

**1 Atmospheric pressure**

**2 The degree of purity of the substance**

**2-What factors impact the evaporation process**

- **Temperature**
- **Surface Area**
- **Humidity**
- **Air current**

### 3-How can you speed up the process of drying wet clothes in winter?

To speed up the drying process, you need to increase the rate of evaporation, which can be done by:

- **Reducing Humidity:** by putting the clothes in an open area
- **Increasing the Surface Area** of the clothes exposed to the surrounding area

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**Compare Between Each of the Following**

| Feature | Melting | Freezing |
|---|---|---|
| **Change of State** | Solid to Liquid | Liquid to Solid |
| **Energy** | Requires gaining Thermal energy (heat) | Requires the loss of heat (thermal energy) |
| **Temperature** | Occurs at the melting point | Occurs at the freezing point (the same as melting point for a substance) |

| Feature | Evaporation | Condensation |
|---|---|---|
| **Change of State** | Liquid to Gas | Gas to Liquid |
| **Energy** | Requires the gaining of heat (thermal energy) | Requires the losing of heat (thermal energy) |
| **Temperature** | Occurs at any temperature below the boiling point | occur at any temperature when the vapor cools |
| **Location** | Occurs only at the surface of the liquid | occur throughout the gas when cooled |

## Weekly assessment (3, 4)

**Complete the following sentences with suitable words:**

1- At normal atmospheric pressure, melting point of ice = **0 °C** while boiling point of water = **100 °C**.

2- The **condensation** process is the reverse of **evaporation** process, and the freezing process is the reverse of the **melting** process.

3- As we go up, atmospheric pressure **decreases** and the boiling point of water **becomes less** than 100 °C.

4- **Physical** change of matter is not accompanied by the formation of new substances, whereas **chemical** change produces a new substance.

5- The greatest kinetic energy of matter particles is in the **gaseous** state, and the least kinetic energy is in the **solid** state.
6- The substance changes from one state to another either by **heating** or **cooling**.
7- During **melting** and **boiling** processes, the temperature of the material remains constant.
8- The two factors that affect the melting and boiling points of a substance are **Atmospheric pressure** and **the degree of purity of the substance**.
9- **Water** boils at a temperature of 100 °C under normal atmospheric pressure, and as atmospheric pressure increases, its boiling point becomes **greater** than 100 °C, and its freezing point becomes **less** than 0 °C.

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**-Write the suitable scientific term for each of the following statements:**

1- The change of matter from the solid state to the gaseous state directly: **Sublimation**
2- The change of matter from the gaseous state to the solid state directly: **Deposition**
3- Coffee dissolves in water faster than regular coffee: **Instant Coffee**
4- A pot cooks food faster because it traps steam inside: **Pressure Cooker**
5- The temperature at which matter changes from a solid to a liquid state: **Melting Point**

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**Give a reason for each of the following:**

**1- Feeling uncomfortable in hot, humid weather.**

- Because in hot weather, sweating increases to alleviate the sensation of heat, but when the humidity level is high, the rate of sweat evaporation decreases, which causes a feeling of discomfort.

- **2- Wet clothes dry faster on sunny days than on humid days.**
- On sunny days, the higher temperature increases the kinetic energy of water molecules, which increases the rate of evaporation. Conversely, on humid days, the air already contains a large amount of water vapor, which decreases the amount of water to escape from the clothes, thus slowing down the rate of evaporation.

**3- The pressure cooker saves fuel.**

- Trapping the steam inside the pressure cooker increases the pressure, which raises the boiling point of water above 100°C. This reduces the cooking time and helps save fuel.

**4- The boiling point of water decreases with increasing altitude above sea level.**

- As the altitude increases above sea level, the atmospheric pressure decreases, and the temperature drops by about 1°C for every 300 m of altitude, which in turn lowers the boiling point of water below 100°C.

### 5- Water evaporates faster from the pot when the stove temperature is raised.

- This is due to the increase in the number of molecules that have enough thermal energy from the stove to escape from the surface of the liquid.

### 6- Clothes dry faster in summer than in winter.

- The higher temperature in the summer increases the number of molecules that have enough thermal energy from the surrounding environment to escape from the surface of the clothes, so they dry faster.

### 7- Wet clothes dry faster when exposed to air currents.

- Because Air currents work to evaporate more molecules from the surface of the water on the clothes, so they dry faster.

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## Compare between:-

**1- Evaporation and boiling in view of (Temperature - place of occurrence - effect of pressure)**

| Feature | Evaporation | Boiling |
|---|---|---|
| **Temperature** | Occurs at **any temperature** below the boiling point. | Occurs only at a **fixed temperature** (the boiling point) . |
| **Place of Occurring** | occurs only to the molecule on the surface of the liquid. | It occurs to all molecules of the liquid. |
| **Effect of Pressure** | The evaporation decreases With higher pressure and increases with lower pressure | the boiling point increases with higher pressure and decreases with lower pressure. |

**2- Evaporation and sublimation, mention an example for each.**

| Feature | Evaporation | Sublimation |
|---|---|---|
| **Phase Transition** | A liquid substance turns Into Gas (Vapor) | Solid substance turns into a Gas (Vapor) directly without turning into a liquid state |
| **Example** | Water is disappearing from clothes on a sunny day. | Dry ice turns directly into gas. |
| | Occurs when the substance gains thermal energy from the surroundings | Occurs when the substance gains thermal energy from the surroundings or by heating |

**Choose the correct answer:-**

**1-In which of the following processes the substance gain heat energy?**

a) Freezing b) Condensation

c) **Sublimation** d) Deposition

2- **When solid dry ice (CO2) is heated, it turns directly into gas without passing through the liquid state. This process is called...**

a) Evaporation b) boiling

c) **sublimation** d) melting

3- **The following table shows a variety of places at different altitudes...**

**-Match each location with the appropriate boiling point of water for that location.**

**What is your conclusion?**

| Boiling point of water | Altitude | Location |
|---|---|---|
| **80.33 °C** | −427 m | 1- Dead Sea |
| **101.4 °C** | 8848 m | 2- Everest Summit |
| **87.71 °C** | 0 m | 3- Sea level |
| **69.94 °C** | 3640 m | 4- La Paz |
| **100.1 °C** | 5859 m | 5- Kilimanjaro |
| **99.96 °C** | 14 m | 6- London |
| **100 °C** | −28 m | 7- Baku |

**Matching:**

| Location | Altitude (m) | Boiling point (°C) |
|---|---|---|
| **1. Dead Sea** | **−427** | **101.4** |
| **2. Everest Summit** | **8848** | **69.94** |
| **3. Sea level** | **0** | **99.96** |
| **4. La Paz** | **3640** | **87.71** |
| **5. Kilimanjaro** | **5859** | **80.33** |
| **6. London** | **14** | **100** |
| **7. Baku** | **−28** | **100.1** |

- 80.33 °C → **Kilimanjaro** (Very high altitude)
- 101.4 °C → **Dead Sea** (Lowest altitude/highest pressure)
- 87.71 °C → **La Paz** (High altitude)
- 69.94 °C → **Everest Summit** (Highest altitude/lowest pressure)

- 100.1 °C → **Baku** (Slightly below sea level)
- 99.96 °C → **London** (Near sea level)
- 100 °C → **Sea level** (0 m is standard sea level)
  **Conclusion:** The boiling point of water is **inversely proportional** to the altitude (and thus the atmospheric pressure).
- As altitude **increases** (pressure decreases), the boiling point **decreases** (e.g., Everest Summit).
- As altitude **decreases** (pressure increases), the boiling point **increases** (e.g., Dead Sea).

**Discuss: -**

**What factors affect the melting and boiling points?**

1- **Atmospheric pressure**

As altitude increases above sea level, the atmospheric pressure decreases below the normal atmospheric pressure. Consequently, • The boiling point decreases, where the temperature decreases by approximately 1°C for every 300 m of altitude, and the Melting point increases and vice versa

2- **The degree of purity of the substance**

Impurities affect the attraction forces between the particles of a pure substance. This leads to a change in the amount of energy needed to change its state. So, its melting and boiling points change. Therefore, the purity of substances is tested by comparing their boiling point or melting point with their fixed known values, in their pure state

The increase in the boiling point and the decrease in the melting point of a solution depend on the concentration of the solution.

**What factors affect the evaporation process?**

1- **Temperature:** The evaporation rate increases with increasing temperature. The evaporation rate increases due to an increase in the number of molecules that have sufficient thermal energy from the surrounding medium to overcome the attractive forces of other molecules and escape from the surface of the liquid into the air.

2- **Surface area of the liquid exposed:** The evaporation rate increases by increasing the surface area of the liquid which exposed to evaporation. The evaporation rate increases due to an increase in the number of molecules that can gain the required thermal energy from the surrounding medium to be released from the surface of the liquid

3- **Air Currents**: The evaporation rate increases with increasing air currents. Air currents increase the evaporation of more water molecules from the surface of the liquid

4- **Humidity:** The evaporation rate increases as decreasing humidity percentage. Trapping water vapor in the air of the container reduces the evaporation rate, while releasing water vapor from the container, which is uncovered, increases the rate of evaporation of more surface molecules

# science

# Performance and assessments answers

# 2nd prep

# Week 5

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# Home performance

## 1-Write the suitable scientific term for each of the following statements:-

1- A system in which there is no exchange of energy and matter with the surrounding medium. (**Isolated system**)
2- A liquid used in cooling due to its high specific heat value. (**Water**)
3- Any part of the universe that is subject to study and observation of changes in energy and matter. (**System**)
4- It is considered a measure of the average kinetic energy of the particles in the system. (**Temperature**)

## 2-Give reason for each of the following:-

**1- Mercury is used in the manufacture of thermometers.**
Because the specific heat of mercury is low and its temperature rises quickly when it absorbs any amount of heat.

**2- Water makes up a large proportion of the human body.**
Because the specific heat of water is high, so it maintains a constant body temperature when the temperature of the surroundings changes

**3- Copper is preferred for use in the manufacture of heating utensils due to its thermal properties.**
Because copper has a low specific heat and its temperature rises quickly when it absorbs any amount of heat, which allows food to cook quickly.

**4- The temperature of ice rises more than that of water when they both acquire the same amount of thermal energy, and their masses are equal.**
Because the specific heat of ice is less than the specific heat of water.

## 3-Answer the following:

**What factors affect the change in the temperature of objects?**
1 The mass of the substance.
2 The type of the substance.
3 The state of the substance.

**Study the corresponding figure and then answer:**



1- The figure shows **Radiator**

2- **Name the liquid used in the figure and explain why.**

The liquid used in this device is water

**Why?** Due to its high specific heat, it absorbs large quantities of thermal energy without a significant increase in its temperature.

## Weekly assessment (5, 6)

**Complete the following statements with suitable words:**

1. The **open** system exchanges energy and matter with the surrounding medium.
2. The **isolated** system does not exchange energy or matter with the surrounding medium.
3. The **closed** system exchanges energy but not matter with the surrounding medium.
4. **Internal energy** is the sum of the kinetic energy and potential energy of all particles of the system.
5. **Temperature** represents the degree of hotness or coldness of the system.

**Choose the correct answer:**

**1- The internal energy of any system is**

a) The potential energy of the particles only

b) The kinetic energy of the particles only

**c) The sum of kinetic energy and potential energy of the particles**

d) The emitted heat

**2- When heating a substance, the kinetic energy of the particles.....**

a) Decreases b) Remains unchanged

**c) Increases** d) Disappears

**3- The change in temperature is calculated by**

a) Electric potential **b) Thermometer**

c) Distance meter d) Barometer

4- A system that does not exchange matter or energy with the surrounding medium is .....

a) An open system b) A closed system

**c) An isolated system** d) A thermal system

**5- When the internal energy of a substance increases,**

a) The temperature decreases **b) The temperature increases**

c) The temperature remains the same d) The particles stop moving

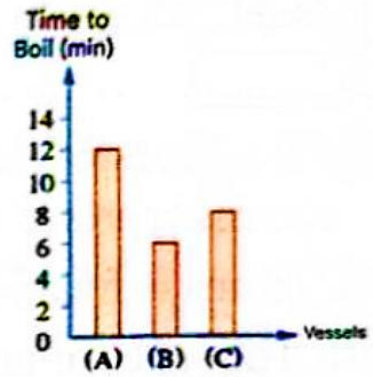
**Does the relative stability of the river water temperature mean that it does not absorb thermal energy? Explain.**

No, the relative stability of the river water temperature does not mean that it does not absorb or release thermal energy.

**Explanation:**

Because Water has a high Specific heat

**The following graph shows the time it takes for three vessels containing different amounts of water to reach the boiling point.**




**a) Which of these vessels contains the largest amount of water?**

Vessel A contains the largest amount of water.

**b) Identify the controlled variable and the independent variable in this experiment.**

- **Controlled Variable(s):**
- -the initial temperature of the water
- -the type of liquid (water)
- -the heat source/rate of heating.
- **Independent Variable**: the amount (mass or volume) of water in the vessel.

---

**What is the difference between internal energy and temperature, according to scientific terms?**

| Term | Scientific Definition |
|---|---|
| **Internal Energy** | The summation of the potential energy and the kinetic energy of the particles of the system |
| **Temperature** | A measure of the average kinetic energy of the particles of the system. |

**Why is the average kinetic energy of hot water particles greater than that of cold water?**

Because an increase in the average kinetic energy of the particles in the system leads to an increase in its temperature, so hot water particles have a higher average kinetic energy than cold water particles.

# science

# October monthly exercises

# 2nd prep

# Week 5

Dr. Science Nova 01070146814

# Exercise No. (1)

**First Question:**

**Complete the following statements with suitable words:**

1- Substances can be classified according to their physical state to **three** common states, that are solid, **liquids**, and gases.

2. Fluids are substances that are in a **liquid** and **gaseous** states.
3. The matter is compressed in the **gaseous** state to reduce **the volume**
4. The forces of attraction between particles of liquid matter are **relatively weak** and the interparticle distances are **relatively large**.
5. There is a fourth state of matter known as **plasma**. It is found in outer space, such as **stars** and **cosmic nebulae**.
6. Plasma is used in air conditioners to purify air by **breaking down** pollutants and germs.
7. **Sublimation** is a reverse process of deposition process.
8. The forces of attraction between particles of solid matter are **very strong** and the interparticle distances are **very small**.
9. Plasma is found in **lightning and the aurora borealis**.
10. The states of matter vary depending on the **forces of attraction** between particles.
11. The substance changes from one state to another either by **heating** (or **thermal energy gain**) or **cooling** (or **thermal energy loss**).
12. At normal atmospheric pressure, the melting point of snow = **0°C** while the boiling point of water = **100°C**.
13. The greatest kinetic energy of matter particles is in the **gaseous** state, and the least kinetic energy is in the **solid** state.
14. The interparticle distances are as large as possible in the **gaseous** state, due to **very weak forces of attraction** between the particles.
15. The interparticle distances are as small as possible in the case of **solid** state, due to **very strong forces of attraction** between the particles.
16. From the factors affecting melting and boiling points: **atmospheric pressure** and **purity of the substance**.
17. The speed of drying clothes can be increased by **increasing the temperature** or **increasing the exposed surface area**.
18. A system is any part of the **universe** that is under study, where changes in **energy** and **matter** are observed.
19. The systems are divided into **open**, **closed**, and **isolated** systems.
20. Change in **temperature** refers to transfer energy from or to the system.

**Write the Suitable Scientific Term for Each of the Following Statements**

1. A state of matter that has the ability to spread and compress. (**Gas state**)
2. A theory that explains the behavior and properties of matter. (**Particle Theory of Matter**)
3. Coffee dissolves in water faster than regular coffee. (**Instant Coffee**.)
4. The temperature at which matter changes from a solid to a liquid state. (**Melting Point**)
5. The transformation of matter from a solid state to a liquid state by raising the temperature. (**Melting**)
6. The transformation of matter from a gaseous state to a liquid state by lowering the temperature. (**Condensation**)
7. The amount of heat required to raise the temperature of 1 kilogram of matter by 1°C. (**Specific Heat**)
8. A system in which there is an exchange of matter and energy with the surrounding environment. **(Open System**)
9. A system in which there is no exchange of matter and energy with the surrounding environment. (**Isolated System**)
10. A system in which there is an exchange of energy without matter with the surrounding environment. (**Closed System**)

---

**Question 3: Give Reasons for:**

1. **Gaseous substances do not have a fixed shape or definite volume.**
   This is because the attractive forces between their particles are very weak, and the spaces between them are very large.

2. **Particles of matter spread faster in gases than in liquids.**
   Because the particles in gases have much greater kinetic energy and larger interparticle distances, their particles move randomly and completely freely in all directions than those in liquids.

3. **It is difficult to break a piece of rock.**
   This is because the attractive forces between the particles of a solid rock are very strong, and the spaces between them are very small, which makes the particles very tightly bound to each other.

4. **Feeling uncomfortable in hot, humid weather.**
   In hot weather, sweating increases to alleviate the sensation of heat, but when the humidity level is high, the rate of sweat evaporation decreases, which causes a feeling of discomfort.

5. **The boiling point of water decreases with altitude above sea level.**
   Because as the altitude increases above sea level, the atmospheric pressure decreases, and the temperature drops by about 1°C for every 300 m of altitude, which in turn lowers the boiling point of water below 100°C.

6. **Clothes dry faster in summer than in winter.**
   The higher temperature in the summer increases the number of molecules that have enough thermal energy from the surrounding environment to escape from the surface of the clothes, so they dry faster.

7. **Copper and aluminum are thermally preferable materials for making heating utensils.**
   Because copper has a low specific heat and its temperature rises quickly when it absorbs any amount of heat, which allows food to cook quickly.

8. **Mercury is used in the manufacture of thermometers.**
   Because the specific heat of mercury is low and its temperature rises quickly when it absorbs any amount of heat.

9. **Water makes up a large proportion of the human body.**
   Because the specific heat of water is high, so it maintains a constant body temperature when the temperature of the surroundings changes.

10. **Water is used to cool many systems.**
    Due to its high specific heat, it absorbs large quantities of thermal energy without a significant increase in its temperature.

**Question 4: Identify the Factors Affecting Each of the Following:**

1. **Speed of diffusion in liquids:**
   - **Temperature:** Higher temperature increases speed.
   - **Mass of the diffusing particles:** Smaller particles diffuse faster.

2. **Melting and boiling points:**
   - **atmospheric Pressure:** as it decreases, the boiling point decreases and the melting point increases.
   - **Purity of the substance:** Impurities generally lower the melting point and raise the boiling point.

3. **Evaporation process:**
   - **Temperature:** Higher temperature increases the rate.
   - **Surface Area of the liquid:** A Larger surface area increases the rate.
   - **Air current:** increasing the air current increases the rate.
   - **Humidity:** Lower humidity (less water vapor in the air) increases the rate.

4. **State of matter:**
   - **Temperature (Kinetic Energy):** Determines the particle movement.
   - **Interparticle Distance:** Affects how closely particles are packed.
   - **Forces of Attraction:** Determines the strength holding particles together.

5. **Change in the temperature of objects:**

   1 -The mass of the substance.

   2- The type of the substance.

   3- The state of the substance.

**Question 5: Define each of the following:**

 **Brownian Motion:** The random motion of relatively large particles suspended in a fluid in all directions as a result of their continuous collisions with the molecules of the fluid.

 **Sublimation:** The change of a substance directly from the solid state to the gaseous state without passing through the liquid state.

 **Particle Theory of Matter:** the theories that have succeeded in explaining much of the behavior and properties of matter

 **Plasma:** The fourth state of matter in which the gas becomes a mixture of positively charged ions and negatively charged free electrons.

 **Precipitation:** the process where water vapor (gas) in the atmosphere condenses and falls to the Earth as rain, snow, sleet, or hail.

Or: it refers to the formation of a solid in a solution.

**Deposition process:** The change of a substance directly from the gaseous state to the solid state without passing through the liquid state.

 **Closed System:** A system in which only energy is exchanged with the surroundings, but matter is not exchanged.

 **Temperature:** A measure of the average kinetic energy of the particles of the system.

# science

# Home work

## 2nd prep

## Week 7

Dr. Science Nova 01070146814

# Lesson (4)

# Method of heat transfer

## Home performance

**1. Explain how the properties of a material affect the rate of heat transfer by conduction.**

The rate of heat transfer by conduction is affected by Thermal Conductivity.
Where Substances can be classified according to their thermal conductivity into:
**1**- Thermal conductors: They are good heat-conducting substances (have high thermal conductivity), like Diamonds. Metals in general, such as aluminum and copper.
**2**- Thermal insulators: They are poor heat-conducting substances (have low thermal conductivity), like Wood. Plastic.

**2. Why does heat not transfer in a vacuum by conduction or convection?**

Because conduction and convection require physical particles to transfer heat.

**3- How does the method of heat transfer differ in metals from that in air?**

- Heat transfer in metals by conduction, where Heat transfers from the object with the highest temperature to the object with the lowest temperature, till thermal equilibrium occurs. It takes place as a result of contact between the particles of the solids.
- While heat transfer in air by convection where when air is heated, its density decreases, and it rises, and it is replaced by the denser cold air.
- Heat can also transfer in air by radiation without the need for material particles.

**4. Explain why:**

**Firefighters wear shiny silver suits.**

Because shiny objects reflect infrared rays well.

**Polystyrene boards are used between hollow bricks when constructing building walls.**

Because polystyrene is a thermal insulator substance that helps to avoid rapid temperature changes inside buildings when temperatures change outside, that leads to reducing air conditioning costs within the buildings.

# Weekly Assessment (7)

**1. Define each of the following:**

| Term | Definition |
|---|---|
| **Thermal Equilibrium** | The state reached by two systems with different temperatures after they come into contact, and their temperatures become equal. |
| **Heat** | The energy that is transferred from one system to another or to the surroundings due to a difference in their temperatures. |
| **Thermal Conduction** | The transfer of thermal energy through solid objects from one point to another without the particles moving from their positions. |
| **Thermally Conductive Materials** | They are good heat-conducting substances (their thermal conductivity is high). |
| **Thermally Insulating Materials** | They are poor heat-conducting substances (their thermal conductivity is low). |

**2. Complete the following sentences with suitable words:**

. Heat transfers in three different ways: **Conduction**, **Convection**, and **Radiation**.
. Energy transfers from a system with a **higher** temperature to a system with a **lower** temperature.
. The sun's heat reaches us by **Radiation**, while a heater's heat reaches us by **Radiation** and **Convection**.
. A **Thermographic** camera can sense the **thermal radiation** emitted by objects and convert it into color images.

**3. Explain why:**

. **Cooking utensils are made of metal, while their handles are made of wood or plastic.**
Because metal is a thermal conductor, while wood and plastic a thermal insulators.

2. **Polystyrene boards are preferred for use between hollow bricks when constructing building walls.**
   - Because polystyrene is a thermal insulator substance that helps to avoid rapid temperature changes inside buildings when temperatures change outside, that leads to a reduction in air conditioning costs within the buildings.
3. **Silver is used in the cooling systems of smartphones.**
   - To get rid of the heat generated within the internal components, which may lead to reduced performance and may damage them.

4. **An eagle soars at high altitudes without flapping its wings.**
   Because the warm air convection currents rising from the Earth's surface cause it to rise upwards, which is equal to the gravitational force, and keeps the eagle floating in the air.
5. **A heater is placed on the floor of a room.**
   - To ensure the distribution of heat throughout the room, where heat is transferred from the heater to the surrounding air, causing the warmer, less dense air to rise to the top while the cooler, denser air moves downwards.
6. **The sea breeze phenomenon occurs.**
   - Due to the fact that land heats up more quickly than water during the day, this is due to the lower specific heat of land compared to that of seawater..
7. **Heat transfers from the sun to the Earth despite the great distance between them.**
   - Because the Sun's heat is transferred to us by radiation.
8. **Heat transfers from a hot body to a cold body.**
   - Because when a body is heated, the average kinetic energy of its particles increases, causing them to vibrate to a greater extent. Some of the energy of these particles is transferred when they collide with neighboring particles, which increases the kinetic energy of these neighboring particles. In the same manner, part of the energy is transferred to the next particles.
9. **The refrigerator freezer is placed at the top of the refrigerator.**
   To ensure heat distribution throughout the refrigerator, as the denser cold air sinks to the bottom and allowing the lighter warm air to rise to the top.
10. **Snakes are able to hunt their prey at night.**
    - Because they receive infrared rays emitted from the bodies of prey through sensory receptors located at the front of their head.

---

**4. Questions**

**1. How does heat transfer in an electric oven? And why?**



Heat transfers by a combination of:

- **Radiation:** The hot heating elements and the hot inner walls of the oven emit **infrared radiation** that travels directly to the food, cooking its surfaces.
- **Convection:** The hot air near the heating elements rises and circulates throughout the oven cavity, transferring heat to the food.

**Conduction:** The food resting directly on the metal baking rack is heated by **conduction** from the hot metal.

## 2. Compare between:

### Convection and Radiation. Mention a practical example for each method.

| Feature | Convection | Radiation |
|---|---|---|
| **Mechanism** | The transfer of thermal energy through fluids with the movement of their particles.<br>When fluid is heated, its density decreases, and it rises, and it is replaced by the denser cold fluid particles (higher density) | Transfer of heat by **electromagnetic waves** (e.g., infrared rays). |
| **Medium Requirement** | **Requires** a material medium (fluid). | **Does not require** a medium; it can travel through a vacuum. |
| **Practical Example** | **The space heater** is placed on the floor of the room. To ensure heat distribution throughout the room, as heat is transferred from the space heater to the air around it, causing the warm, lighter air (with lower density) to rise to the top and the colder, denser air (with higher density) to sink. | **Thermography**: A technique that uses a camera to detect the thermal radiation (infrared rays) emitted from objects and convert it into colored images; the colors in these images depend on the changes in the object's temperature. |

# science

## Home work

## &

## Weekly assessment

## 2nd prep

## Week 8, 9

Dr. Science Nova 01070146814

# Unit 2

# Lesson 1

# Chemical reactions

## Home performance

**Assisted with the schoolbook and, in light of what you have studied, explain the meaning of the following words:**

- **Physical change:** A change in the state or shape of a substance without changing its chemical composition.
- **Chemical change:** A change that results in new substances with chemical properties different from those of the original substances.
- **Chemical reaction:** The process of breaking the bonds between atoms of the reacting substances (reactants) and forming new bonds between atoms of the produced substances (products).

---

**Compare between physical and chemical changes? In terms of definition - Examples**

| Feature | Physical Change | Chemical Change |
|---|---|---|
| **Definition** | A change in the state or shape of a substance without changing its chemical composition. | A change that results in new substances with chemical properties different from those of the original substances. |
| **Examples** | • Dissolution of salt in water.<br>• Melting of ice.<br>• Wood cutting.<br>• Water evaporation | • Conversion of milk into yogurt.<br>• Rusting of an iron nail.<br>• Burning a piece of bread in the oven.<br>• Limewater turns milky (turbid) when carbon dioxide gas passes through it.<br>• The color of a damp (wet with water) universal indicator strip is changed when it is brought close to ammonia gas. |

## What is meant by a chemical reaction?

- The process of breaking the bonds between atoms of the reacting substances (reactants) and forming new bonds between atoms of the produced substances (products).

---

## What happens when:

### 1. Adding a silver nitrate solution to a sodium chloride solution.

A white precipitate of silver chloride is formed, which turns violet when exposed to sunlight.

---

### 2. Exposing a silver chloride precipitate to sunlight.

Its color is turning violet.

---

### 3. Placing a zinc plate in a copper sulphate solution.

The blue color of the copper sulphate solution disappears, and reddish-brown copper is deposited over the zinc plate.

---

### 4. Placing a magnesium strip in a test tube containing dilute hydrochloric acid.

Formation of hydrogen gas $H_2$, bubbles.

---

### 5. Lighting a magnesium ribbon in the air.

- Emission of light and heat, where it produces a glow of the strip with a bright light accompanied by heat emission.
- The strip converts into a white powder of magnesium oxide, MgO, which is soluble in water.

---

### 6. Heating a spoon with sugar on a flame for a period of time.

- Sugar gradually converts into a brown molten substance.
- Strong odour appears (caramel smell).

• After complete combustion:

- Evolving of smoke.
- The colour changes.
- A solid black substance (carbon) remains.

---

### 7. Adding sodium hydroxide to cooking oil.

- a change occurring in the consistency of the mixture over time, then it turns into a new solid substance.

**How do we infer the occurrence of a chemical reaction?**

- Formation of a Precipitate
- Evolution of a Gas
- Change in Color
- Emission of Light or heat
- Emission of odour
- Change in consistency

---

**Choose from column (B) the suitable formula from column (A):**

| Element/Compound | Chemical formula or symbol |
|---|---|
| 1- **Silver nitrate** | **$AgNO_3$** |
| 2- **Silver Chloride** | **AgCl** |
| 3- **Sodium Chloride** | **NaCl** |
| 4- **Copper Sulphate** | **$CuSO_4$** |
| 5- **Zinc Sulphate** | **$ZnSO_4$** |
| 6- **Hydrochloric acid** | **HCl** |
| 7- **Magnesium** | **Mg** |
| 8- **Copper** | **Cu** |
| 9- **Hydrogen** | **$H_2$** |
| 10- **Magnesium Chloride** | **$MgCl_2$** |
| 11- **Magnesium Oxide** | **MgO** |
| 12- **Sulphuric acid** | **$H_2SO_4$** |
| 13- **Sodium bicarbonate (Baking soda)** | **$NaHCO_3$** |
| 14- **Sodium hydroxide (Caustic Soda)** | **NaOH** |

# Weekly assessment

# 8, 9

**What is meant by each of the following:**

- **Physical change:** A change in the state or shape of a substance without changing its chemical composition.
- **Chemical change:** A change that results in new substances with chemical properties different from those of the original substances.
- **Chemical reaction:** The process of breaking the bonds between atoms of the reacting substances (reactants) and forming new bonds between atoms of the produced substances (products).

---

**Give a scientific reason for each of the following:**

1. **The melting of a candle is considered a physical change, while the burning of coal is a chemical change.**
   - Because melting a candle changes only the state of the wax, the wax is chemically the same. Burning coal involves a reaction that forms a new substance with different chemical properties.

---

2. **When a zinc plate is placed in a beaker containing a blue copper sulphate solution, the blue color disappears.**
   - due to the formation of the colourless zinc sulphate solution $ZnSO_4$.

---

3. **A magnesium strip must be held with tongs before lighting it with a flame.**
   - to avoid burns during its combustion.

---

**What do you observe when...?**

1. **Burning table sugar**
   A solid black substance (carbon) is formed, with smoke and a strong odour.

---

2. **Heating a mixture of caustic soda and oil with continuous stirring**
   The liquid consistency changes, and solid soap is formed.

---

3. **Slices of apple or potato are left exposed to the open air**
   The colour of apple or potato slices turns brown.

**How do you infer that a chemical reaction has occurred in the following experiments?**

1. **Frying an egg:** Both the colour and texture of the white part (egg whites) and yellow part (egg yolk) of eggs change.
2. **Igniting fireworks:** Emission of light and heat.
3. **Adding a piece of a magnesium strip to a test tube containing dilute hydrochloric acid:** Formation of hydrogen gas $H_2$ bubbles.

---

**Write the chemical formula for the following compounds:**

| Compound | Chemical formula |
|---|---|
| 1- Silver nitrate | $AgNO_3$ |
| 2- Silver Chloride | AgCl |
| 3- Sodium Chloride | NaCl |
| 4- Copper Sulphate | $CuSO_4$ |
| 5- Zinc Sulphate | $ZnSO_4$ |
| 6- Hydrochloric acid | HCl |
| 7- Magnesium Oxide | MgO |
| 8- Sodium hydroxide | NaOH |
| 9- Sodium nitrate | $NaNO_3$ |
| 10- Copper chloride | $CuCl_2$ |

science
prep 2
1ST term

# November Revision Answers

**(1)What is meant by each of the following?**

1- System :Any part of the universe that is under study where the changes in energy and matter are observed.

2- Open system :A system in which both energy and matter are exchanged with the surroundings.

3- Closed system: A system in which only energy is exchanged with the surroundings, but matter is not exchanged.

4- Isolated system: system in which neither energy nor matter is exchanged with the surroundings.

5- Internal Energy of the system: The summation of the potential energy and the kinetic energy of the particles of the system.

6- Temperature of the system : A measure of the average kinetic energy of the particles of the system.

7- Heat : The energy that is transferred from one system to another or to the surroundings due to a difference in their temperatures.

8- Specific Heat : The quantity of heat required to raise the temperature of 1 kg of a substance by 1°C

9- Thermal equilibrium : The state reached by two systems with different temperatures after they come into contact and their temperatures become equal.

10- Thermal conduction :The transfer of thermal energy through solid objects from one point to another without the particles moving from their positions.

11- Thermal conductivity: The measure for the extent to which a substance can conduct heat through it.

12- Thermally conductive materials :They are good heat-cconducting substances (their thermal conductivity is high).

13-Thermally insulating Materials:They are poor heat-conducting substances (their thermal conductivity is low).

14- Sea breeze :Cool air (breeze) blowing from the sea towards the land during the daytime.

15.Thermal radiation The transfer of infrared radiation from the surfaces of objects, especially hot objects, without the need for material particles.

16- Physical change: A change in the state or shape of a substance without changing its chemical composition.

17- chemical change: A change that results in new substances with chemical properties different from those of the original substances.

18- Chemical equation: The process of breaking the bonds between atoms of the reacting Chemical reaction substances (reactants) and forming new bonds between atoms of the produced substances (products).

19- Balanced chemical equation: It is a symbolic representation of the chemical reaction, showing the reactants and the products by using symbols and Symbolic equation molecular formulas, with the simplest integer ratio between the numbers of atoms or molecules entering and resulting from the reaction.

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20- Catalysts: A substance that increases the speed of the reaction without being consumed or changed.

21- Law of Conservation of Mass: The total mass of the substances involved in the chemical reaction is equal to the total mass of the substances resulting from it.

22 - Coefficients :The numbers that precede the symbols of the elements or the molecular formulas of the compounds in the balanced symbolic equation.

23-Subscripts : The numbers located below the symbols of the elements that make up the molecule and represent the number of atoms in a single molecule.

24-Nutritional chemistry:A branch of chemistry that studies the types of nutrients in food

25-Monosaccharides :The simplest form of carbohydrates, they cannot be decomposed (by water) into simpler substances.

26-Disaccharides : They are oligosaccharides, each molecule of which consists of units of monosaccharides with one water molecule removed.

27-Cholesterol : A faty substance whose deposition on the inside wals of arteries, leads to their blockage.

28-Amino acids : The basic units that make up proteins are soluble in water.

**(2)Choose the correct answer for the questions :**

1-The internal energy of water increases when……

a- the temperature of water changes from 70°C to 60°C

b- water vapour condenses on a plant leaf.

C- a quantity of water is heated from 20°C to 30°C

d it is placed in a refrigerator.

2-What are the tools used to find the specific heat of a substance?

a-Balance and stopwatch. b Balance and thermometer.

c - Stopwatch and thermometer. d-Balance, stopwatch and thermometer.

3- Which of the following expresses a chomical change?

(a) Iodine sublimation b ) The conversion of charcoal to ash.

c ) The heating of an iron rod. (d) The molting of ice.

4-A lemon drink its temperature is 20°C was placed with a cube of ice, and after two minutes, the temperature of the drink became 8°C, the temperature of the remaining part of the ice cube was ....

a- 0°C b -2°C c- 4°C d-8°C

5-What happens to the cold air outside a kettle when it comes into contact with its hot outer surface?

a- Its density decreases and it sinks downwards.

b- Its density decreases and rises upwards.

c- Its density increases and it sinks downwards.

d- It s density increases and it rises upwards.

6-The combustion (burning) of table sugar is indicated by the formation .....

a) a solid matter and evolution of smoke.

b) a white precipitate and evolution of smoko.

c) green solution and reddish brown gas.

d) white powder and bright light.

7-The following reactions were carried out:

(1) Burning a strip of magnesium.

(2) Adding hydrochloric acid to magnesium.

(3) Adding sulphuric acid to baking soda.

Which of these reactions is accompanied by the release of gas?

a (1) Only. (b) (2) Only. (C) (1) , (3). d) (2) , (3).

8-Soap is formed by the reaction of……..

(a) oil with solid sodium hydroxide. b- oil with sodium hydroxide solution.

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(c) oil with hydrochloric acid. d ) oil with silver nitrate solution.

9-All chemical reactions are accompanied by the following, exept.....

a- breaking down the bonds between the molecules of the reactants.

b- forming bonds between the molecules of the products.

c- the formation of new substances.

d- the release of heat.

10-In the balanced equation:

$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

What is the total number of coefficients of reactants molecules and the total number of subscripts of the products respectively?

a) 5,7 b) 5,10 c ) 9,5 d)6,5

11-Which of the following expresses the physical state of the substances of the reaction between magnesium metal and hydrochloric acid to form magnesium chloride solution and hydrogen gas?

ⓐ $Mg_{(g)} + HCl_{(\ell)} \longrightarrow MgCl_{2(aq)} + H_{2(g)}$

ⓑ $Mg_{(g)} + HCl_{(aq)} \longrightarrow MgCl_{2(\ell)} + H_{2(g)}$

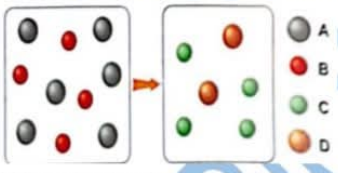
ⓒ $Mg_{(s)} + HCl_{(\ell)} \longrightarrow MgCl_{2(\ell)} + H_{2(\ell)}$

ⓓ $Mg_{(s)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + H_{2(g)}$

12-Which of the following verifies the law of conservation of mass when magnesium reacts with oxygen to form magnesium oxide?

| Choices | MgO | $O_2$ | Mg |
|---|---|---|---|
| ⓐ | 15.36 g | 3.2 g | 4.8 g |
| ⓑ | 1.6 g | 4.8 g | 3.2 g |
| ⓒ | 8 g | 3.2 g | 4.8 g |
| ⓓ | 7.4 g | 4.8 g | 3.2 g |

13-The folowing figure illustrates a chemical reaction :




What is the balanced symbolic equation which represents this reaction?

a) 6A +4B → 4C + 2D

b ) 6A + 4B → C + D

c ) A + 2B → 2C + D

d) 3A + 2B → 2 C +D

**(3)Various Questions:**

1-What are the factors that affect the temperature of a body (system)?

- Mass of matter.
- Type of matter.
- State of matter.

2-What are the steps of a chemical reaction?

step one:Breaking the bonds between the atoms of the molecules of the reactants.

step two:Atoms are rearranged and combined together with new bonds to form the products

3-What are the laboratory safety precautions when burning a magnesium ribbon?

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1. Hold the magnesium strip with heat-resistant tongs, to avoid burns during its combustion.
2. Use a protective mask to avoid damage from the bright light.
3. Do not touch the formed white powder immediately after the reaction, as it will be hot.

4- Mention examples of some chemical reactions that occur in daily life.

- Frying eggs.
- Food spoiling outside the refrigerator.
- Ignition of fireworks.
- Conversion of milk into yogurt.
- Rusting of an iron nail.

**( 4)Compare between each of the following in terms of definition and examples:**

**1- Open systems, closed systems, and isolated systems.**

| ① | The open system | The closed system | The isolated system |
|---|---|---|---|
| Definition | A system in which both energy and matter are exchanged with the surroundings | A system in which only energy is exchanged but the matter is not exchanged with the surroundings | A system in which neither energy nor matter is exchanged with the surroundings |
| Example | An open container with boiling water | A can of soft drink placed in ice | Thermos contain a hot drink |

**2- Thermally conductive materials and thermally insulating materials.**

| ② | Thermal conductors | Thermal insulators |
|---|---|---|
| | They are good heat-conducting substances (high thermal conductivity). | They are poor heat-conducting substances (low thermal conductivity) |
| Example : | • Diamond.<br>• Metals in general, such as: aluminum and copper. | • Wood.<br>• Plastic. |

**(5)Give reason for each of the following:**

**1-Increasing temperature of a substance increases by decreasing its mass when it gains the same amount of thermal energy.**

Because the thermal energy gained by the substance is distributed over fewer number of particles so their average kinetic energy increases to a greater extent.

**2-When heating two equal quantities of water and oil for the same period of time and from the same heat source, the increase in the oil's temperature is greater than that of water.**

Because the specific heat of oil is less than the specific heat of water.

**3- It is preferable to make cooking utensils from copper instead of aluminum.**

Due to its high thermal conductivity

**4- Mercury is used in the manufacture of thermometers.**

Because the specific heat of mercury is low and its temperature rises quickly when it absorbs any amount of heat.

**5- Water is the main component of the human body.**

Because the specific heat of water is high, so it maintains a constant body temperature when the temperature of the surroundings changes.

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**6- When a quantity of iron balls at 80°C is mixed with a quantity of water at 30°C, the temperature of the mixture at equilibrium is not 55°C.**

Because the specific heat of water is greater than the specific heat of iron.

**7-The concept of heat differs from the concept of temperature.**

Because heat is the energy that is transferred from one system to another or to the surroundings due to differences in their temperatures, while temperature is A measure of the average kinetic energy of the particles of the system.

**8- Cooking utensils are made of aluminum and their handles are made of wood.**

Beacause aluminium have high thermal conductivity, while wood has low thermal conductivity.

**9-Polystyrene boards are preferred for use between hollow bricks when constructing buildings.**

Because polystyrene and the air in the brick cavities are heat insulators,which prevent rapid change in temperature inside the buildings when the outside temperature changes

**10-There are cooling systems inside computers that use materials with good thermal conductivity, such as silver.**

To dissipate (get rid of) the excessive heat generated in the internal components which may lead to poor performance or even damage.

**11-A heater is placed on the floor of a room.**

because the warm lighter air (with lower density) rise up to the top of the room and the colder denser air (with higher density) to sink,to ensure heat distribution

**12-The sun's heat reaches us by radiation.**

Because heat transfer by radiation does not require the presence of material particles

**13-Snakes are able to hunt their prey at night.**

Because they receive infrared rays emitted from the bodies of preys through sensory receptors located at the front of their head.

**14- The burning of wood is a chemical change, while the melting of ice is a physical change.**

Because a new substance is formed ,which has different properties from the properties of wood. while melting ice is not accompanied by a change in chemical composition.

**15-When burning a magnesium ribbon, it is a must to wear a protective mask.**

To avoid the damage from the bright light.

**16-The chemical equation must be balanced.**

To verify the law of conservation of matter (conservation of mass).

**17-It is better to express a chemical reaction by using a balanced symbolic equation than a word equation.**

Because the balanced symbolic equation shows the type and number of atoms of the elements involved in the composition of both the reactants and the products.

**18- Balancing the equation: $H_2 + O_2 \rightarrow H_2O$ in this way: $H_2 + O_2 \rightarrow H_2O_2$ is a scientific error.**

Because the balancing should be accomplished by changing the coefficients and not the subscripts.

**(6)Choose from column (B) the suitable for column ( B):**

| | (A) The compound | (B) Chemical formula or symbol |
|---|---|---|
| 1 | Sodium nitrate | HI |
| 2 | Hydrogen iodide | $FeSO_4$ |
| 3 | Iron sulphate | $NH_3$ |
| 4 | Ammonia | $NaNO_3$ |

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**(7)Write the balanced symbolic chemical equations that express the following reactions, with the physical state of the reactants and products:**

1- Reaction of magnesium metal with dilute hydrochloric acid.

$$Mg_{(s)} + 2HCl_{(aq)} \xrightarrow{dil.} MgCl_{2(aq)} + H_{2(g)}$$

2- Combustion of ethane gas (C2H6) in an abundance of oxygen gas, with the reaction condition.

$$2C_2H_{6(g)} + 7O_{2(g)} \xrightarrow{\Delta} 4CO_{2(g)} + 6H_2O_{(l)}$$

3-Heating mercury (II) oxide powder strongly to form mercury and oxygen.

$$2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)$$

4-Reaction of magnesium with dilute sulfuric acid to form magnesium sulfate soluble in water and hydrogen gas.

$$Mg(s) + H_2SO_4(aq) \xrightarrow{dil} MgSO_4(aq) + H_2(g).$$

5-Reaction of solid iodine with hydrogen gas to form hydrogen iodide gas.

$$I_{2(S)} + H_{2(g)} \longrightarrow 2HI_{(g)}$$

6-Reaction of nitrogen gas with hydrogen gas to form ammonia gas.

$$N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$$

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